T3000 API & Protocol

Trend Logging API · built Nov 13, 2026

Trend Logging API

Overview

The Trend Logging API manages historical data collection and retrieval. Configure trendlogs to record sensor values, control outputs, and system performance over time.

Common Operations

Get All Trendlogs

TYPESCRIPT

```
const trendlogs = await fetch('http://localhost:9103/api/t3_device/devices/1/trendlogs')
  .then(res => res.json());
```

Get Trendlog Data

TYPESCRIPT

```
const data = await fetch('http://localhost:9103/api/t3_device/trendlogs/1/data?start=202
  .then(res => res.json());

console.log(data);
// [
//   { timestamp: "2026-01-09T08:00:00Z", value: 72.5 },
//   { timestamp: "2026-01-09T08:05:00Z", value: 72.3 }
// ]
```

Update Trendlog Configuration

```typescript
await fetch('http://localhost:9103/api/t3_device/trendlogs/1', {
  method: 'PUT',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    sample_rate: 300,  // 5 minutes
    enabled: true
  })
});
```

Trendlog Endpoints

GET /api/t3_device/devices/:id/trendlogs

Description: Get all trendlogs for a device

Parameters:

- `id` (path, integer) - Device ID

Request:

```http
GET /api/t3_device/devices/1/trendlogs HTTP/1.1
```

Response:

JSON

```
[
  {
    "id": 1,
    "device_id": 1,
    "panel": 5,
    "index": 1,
    "label": "Room Temp Log",
    "input_index": 1,
    "sample_rate": 300,
    "buffer_size": 1000,
    "enabled": true,
    "status": "RUNNING"
  }
]
```

Used By: TrendlogTable.tsx

GET /api/t3_device/trendlogs/:id/data

Description: Get historical data from trendlog

Parameters:

- `id` (path, integer) - Trendlog ID
- `start` (query, string) - Start date (ISO 8601)
- `end` (query, string) - End date (ISO 8601)
- `limit` (query, integer, optional) - Max records (default: 1000)

Request:

HTTP

```
GET /api/t3_device/trendlogs/1/data?start=2026-01-01T00:00:00Z&end=2026-01-09T23:59:59Z&
```

Response:

JSON


```
{
  "trendlog_id": 1,
  "label": "Room Temp Log",
  "start": "2026-01-01T00:00:00Z",
  "end": "2026-01-09T23:59:59Z",
  "total_records": 2592,
  "data": [
    {
      "timestamp": "2026-01-09T08:00:00Z",
      "value": 72.5,
      "quality": "GOOD"
    },
    {
      "timestamp": "2026-01-09T08:05:00Z",
      "value": 72.3,
      "quality": "GOOD"
    }
  ]
}
```


Used By: TrendChart.tsx

Errors:

- `404` - Trendlog not found
- `400` - Invalid date range

PUT /api/t3_device/trendlogs/:id

Description: Update trendlog configuration

Parameters:

- `id` (path, integer) - Trendlog ID

Request:

```HTTP
PUT /api/t3_device/trendlogs/1 HTTP/1.1
Content-Type: application/json

{
  "label": "Zone 1 Temperature",
  "sample_rate": 600,
  "enabled": true
}
```

Response:

```JSON
{
  "success": true,
  "id": 1
}
```

Used By: EditTrendlogDialog.tsx

POST /api/t3_device/trendlogs/:id/clear

Description: Clear all data from trendlog

Parameters:

- `id` (path, integer) - Trendlog ID

Request:

```HTTP
POST /api/t3_device/trendlogs/1/clear HTTP/1.1
```

Response:

```JSON
{
  "success": true,
  "records_deleted": 2592
}
```

Used By: ClearTrendlogButton.tsx

GET /api/t3_device/trendlogs/:id/export

Description: Export trendlog data as CSV

Parameters:

- `id` (path, integer) - Trendlog ID
- `start` (query, string) - Start date
- `end` (query, string) - End date

Request:

```HTTP
GET /api/t3_device/trendlogs/1/export?start=2026-01-01&end=2026-01-09 HTTP/1.1
```

Response:

```CSV
timestamp,value,quality
2026-01-09T08:00:00Z,72.5,GOOD
2026-01-09T08:05:00Z,72.3,GOOD
```

Used By: ExportTrendlogButton.tsx

Database Schema

SQL

```sql
CREATE TABLE TRENDLOGS (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    device_id INTEGER NOT NULL,
    panel INTEGER NOT NULL,
    index_ INTEGER NOT NULL,
    label TEXT,
    input_index INTEGER,
    sample_rate INTEGER DEFAULT 300,
    buffer_size INTEGER DEFAULT 1000,
    enabled INTEGER DEFAULT 1,
    status TEXT,
    FOREIGN KEY (device_id) REFERENCES DEVICES(id) ON DELETE CASCADE
);

CREATE TABLE TRENDLOG_DATA (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    trendlog_id INTEGER NOT NULL,
    timestamp DATETIME NOT NULL,
    value REAL,
    quality TEXT DEFAULT 'GOOD',
    FOREIGN KEY (trendlog_id) REFERENCES TRENDLOGS(id) ON DELETE CASCADE
);

CREATE INDEX idx_trendlog_data_timestamp ON TRENDLOG_DATA(trendlog_id, timestamp);
```

TypeScript Types

TYPESCRIPT

```typescript
export interface Trendlog {
  id: number;
  device_id: number;
  panel: number;
  index: number;
  label: string;
  input_index: number;
  sample_rate: number;
  buffer_size: number;
  enabled: boolean;
  status: string;
}

export interface TrendlogDataPoint {
  timestamp: string;
  value: number;
  quality: 'GOOD' | 'BAD' | 'UNCERTAIN';
}
```